sectors. In Andhra Pradesh, Hyderabad-based ASIP Technologies has joined forces with South Korea's APACT to launch a packaging and assembly facility for the electronics sector. And in Mohali, Punjab, CDIL has received approval for brownfield expansion of its high-power silicon and SiC device manufacturing for electric vehicles.

THE NEXT STEP

Ankush Wadhera, BCG's Managing Director and India Practice Leader for Semiconductors, gives the government a double tick for the mission's execution so far. Unlike in the past, it has drawn more credible applications than the money it has set aside. "This has made boardrooms across the world in the chip business sit up and ask: Do I need an India strategy? Am I missing out on a big opportunity in India?," says Wadhera. Pankaj Gulati, an industry veteran and president, CDIL, stops only slightly short of embracing his enthusiasm wholeheartedly. "Failure is not an option for any of us," he says. "We must show the world that India can and will do it. But it's important that the government continue to support us for the next four years."

Wadhera, meanwhile, says that while ISM 1.0 may take a decade to bear fruit, India should not lose momentum and must prepare for ISM 2.0. It should focus on three broad areas—a self-reliant input ecosystem of chemicals, gases and substrates; expanding compound semiconductor fabs for use in diverse Indian industries; and establishing advanced fabs to bolster domestic R&D and keep pace with giants like TSMC.

Raja Manickam, founder and CEO of iVP semi and former CEO of Tata Electronics OSAT, says the government also needs to help semiconductor start-ups get the sales side right, including market strategies. "Many of these chip companies are being set up not so much to meet Indian demand but for global needs," he says. "We can tell these and other companies selling us chips to start using the ecosystem being set up and make chips in India itself." CDIL's Gulati echoes the need for focus, urging India to concentrate on end-product markets like automotive and power and pick up world leadership in these rather than spread its resources across a wide spectrum.



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> *Founder & CEO, iVP Semi*

THE TALENT GAP

While exponential expansion of chip-making facilities has got the wheels moving, they could come to a grinding halt without a steady pipeline of trained manpower, especially mid-level managers. BCG's Wadhera doubts if a generalised 'spray and play' approach to skill development will work. What's needed are talent corridors between India and Taiwan or India and Japan for seamless workforce exchange. Deloitte's Thandavarayan believes that while advanced semiconductor manufacturing can be included in the curricula of universities and technical institutes, the government should also establish centres of excellence in collaboration with global academic institutions and chip companies. In a promising move, the government is encouraging young talent by using the integrated semiconductor facilities in the public sector SCL. After start-ups and university students design them, SCL fabricates these chips as proofs of concept to validate them.

SCL itself is stepping up to become the backbone of India's strategic sectors, especially space, defence and atomic energy. "What the strategic sector needs is reliability," says SCL director-general Dr Kamaljeet Singh, "as there is a chip war going on, and we could be vulnerable. The chips we make for ISRO are fully indigenous, from fabrication to packaging. What India should also develop is a clear roadmap of what we require and become a leader in these segments rather than trying to develop expertise in a vast area. We should not try to be *shaane* (oversmart), but proceed step by step."

That's good advice. If the Modi government's big bet on chips pays off, India could meet 35 per cent of its needs domestically within a decade. That's huge. The pace of indigenisation is expected to accelerate as the ecosystem stabilises. At the launch of the mission, Modi had said, "Our dream is that every device in the world will have an Indian-made chip." It's an ambition worth pursuing. ■



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>
> **DR KAMALJEET SINGH** | *Director General, SCL*